

Tutorial 1: Exercise on Structure of a Program

1. Type the following code in your C++ editor and then compile and run the program after saving the file.

```
#include<iostream>
int sample()
{
    std::cout<<"Main function should be included in any C++
program";
    return 0;
}
```

What is the output of the program?

Refer to the error message, re run the program after fixing the error.

2. Write a C++ program that prints out the following menu for a game (include the line of asterisks (*) on top and bottom):

```
*****
```

Welcome!

Please choose a number from the following options:

1. Play the game!
2. Demo the game!
3. Exit

```
*****
```

2. Write a C++ program that will calculate the average of four numbers that are stored in variables. The variables are all of the data type: double. The values that are stored are 578, 986, 1066, and 714.

Display a message showing the sum of all four numbers

(ex. The sum of those numbers is ____.)

On the next line, display the average of all four numbers

(ex. – “The average of those numbers is ____.”)

3. Write a C++ program that calculates the average of four numbers – just like in the previous exercise – but this time ask the user to enter which four numbers to calculate. You will have to use the cin command to get all four numbers from the user.

Sample Output:

Enter the first number: (number1)

Enter the second number: (number2)

Enter the third number: (number3)

Enter the fourth number: (number4)

The average of these numbers is: (average)

4. Write a C++ program that will calculate how much of a profit an investor will make. You will need to get the following three inputs:

- the number of shares the investor purchased
- the price of the stock (per share) when the investor purchased it
- the price of the stock (per share) now

The formula to use: Profit = (number of shares * Current Price) – (number of shares * Purchase Price)

Output this message to the screen: You have made a profit of \$_____ dollars since you bought _____ shares of this stock.